PART II: KINETIC THEORY OF GASES

Exercise 1.

In solving the ideal-gas law equation $pV = nRT$ for n, we first convert the temperature to the Kelvin scale: T_i = 313.15 K, and the volume to SI units: $V_i = 10^{-3}$ m^3.

(a) The number of moles of oxygen present is

$$n = \frac{pV_i}{RT_i} = \frac{(1.01\times10^5\ \text{Pa})(1.000\times10^{-3}\ \text{m}^3)}{(8.31\text{J/mol}\cdot\text{K})(313.15\text{K})} = 3.88\times10^{-2}\ \text{mol}.$$

(b) Similarly, the ideal gas law $pV = nRT$ leads to

$$T_f = \frac{pV_f}{nR} = \frac{(1.06\times10^5\ \text{Pa})(1.500\times10^{-3}\ \text{m}^3)}{(3.88\times10^{-2}\ \text{mol})(8.31\text{J/mol}\cdot\text{K})} = 493\text{K}.$$

Exercise 2.

(a) With T = 283 K, we obtain

$$n = \frac{pV}{RT} = \frac{100\times10^3\ \text{Pa}\quad 2.50\,\text{m}^3}{8.31\text{J/mol}\cdot\text{K}\quad 283\text{K}} = 106\,\text{mol}.$$

(b) We can use the answer to part (a) with the new values of pressure and temperature, and solve the ideal gas law for the new volume, or we could set up the gas law in ratio form as:

$$\frac{p_fV_f}{p_iV_i} = \frac{T_f}{T_i}$$

(where $n_i = n_f$ and thus cancels out), which yields a final volume of

$$V_f = V_i\left(\frac{p_i}{p_f}\right)\left(\frac{T_f}{T_i}\right) = (2.50\,\text{m}^3)\left(\frac{100\,\text{kPa}}{300\,\text{kPa}}\right)\left(\frac{303\,\text{K}}{283\,\text{K}}\right) = 0.892\ \text{m}^3.$$

Exercise 3.

With V = 1.0 x 10^{-6} m^3, p = 1.01 x 10^{-13} Pa, and T = 293 K, the ideal gas law gives

$$n = \frac{pV}{RT} = \frac{(1.01\times10^{-13}\ \text{Pa})\,(1.0\times10^{-6}\ \text{m}^3)}{(8.31\ \text{J/mol}\cdot\text{K})(293\ \text{K})} = 4.1\times10^{-23}\ \text{mole}.$$

Subsequently, with the Avogadro's number N_A, we have $N = nN_A$ = 25 molecules. We can express this as a ratio (with V now written as 1 cm^3) N/V = 25 molecules/cm^3.

Exercise 4.

(a) We have $\Delta E_{\text{int}} = nC_V\Delta T$ (which gives 0) implying $Q = W$.

Then, apply the equation

$$W = nRT\ln\frac{V_f}{V_i}$$

with T = 303 K, Q = -3.14 x 10^3 J

(b) That negative sign in the result of part (a) implies the transfer of heat is from the gas.

Exercise 5.

(a) At the surface, the air volume is

$$V_1 = Ah = \pi(1.00\text{ m})^2(4.00\text{ m}) = 12.57\text{ m}^3 \approx 12.6\text{ m}^3.$$

(b) The temperature and pressure of the air inside the submarine at the surface are T_1 = 20 °C = 293 K and $p_1 = p_0$ = 1.00 atm. On the other hand, at depth h = 80 m, we have T_2 = -30 °C = 243 K and

$$p_2 = p_0 + \rho gh = 1.00\text{ atm} + (1024\text{ kg/m}^3)(9.80\text{ m/s}^2)(80.0\text{ m})\frac{1.00\text{ atm}}{1.01\times10^5\text{ Pa}}$$

$$= 1.00\text{ atm} + 7.95\text{ atm} = 8.95\text{ atm}.$$

Atm + $(kg/m^3)(m/s^2)(m)$ =Atm + kg/ms^2 = Atm + Pascal

(Pascal = N/m^2= J/m^3 = kg/ms^2, 1 atm = 10^5 Pascal)

Therefore, using the ideal gas law, $pV = NkT$, the air volume at this depth would be

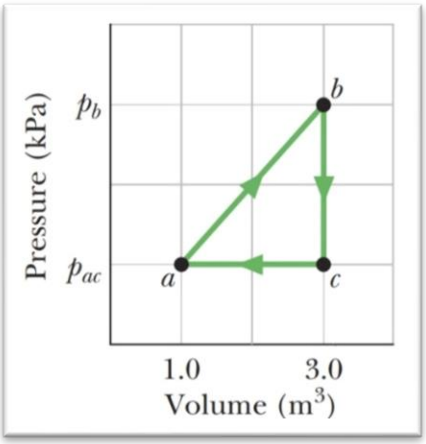
$$\frac{p_1V_1}{p_2V_2} = \frac{T_1}{T_2} \quad\Rightarrow\quad V_2 = \left(\frac{p_1}{p_2}\right)\left(\frac{T_2}{T_1}\right)V_1 = \left(\frac{1.00\text{ atm}}{8.95\text{ atm}}\right)\left(\frac{243\text{ K}}{293\text{ K}}\right)(12.57\text{ m}^3) = 1.16\text{ m}^3.$$

(c) The decrease in volume is $\Delta V = V_1 - V_2$ = 11.44 m^3. Using the ideal gas law, the amount of air this volume corresponds to is

$$n = \frac{p\Delta V}{RT} = \frac{(8.95\text{ atm})(1.01\times10^5\text{ Pa/atm})(11.44\text{ m}^3)}{(8.31\text{ J/mol}\cdot\text{K})(243\text{ K})} = 5.10\times10^3\text{ mol}.$$

Thus, in order for the submarine to maintain the original air volume in the chamber, 5.10x10^3 mol of air must be released.

Exercise 6.



(a) At point a, we know enough information to compute n:

$$n = \frac{pV}{RT} = \frac{(2500\,\text{Pa})(1.0\,\text{m}^3)}{(8.31\,\text{J/mol}\cdot\text{K})(200\,\text{K})} = 1.5\,\text{mol}.$$

(b) We can use the answer to part (a) with the new values of pressure and volume, and solve the ideal gas law for the new temperature, or we could set up the gas law in terms of ratios (note: $n_a = n_b$ and cancels out):

$$\frac{p_b V_b}{p_a V_a} = \frac{T_b}{T_a} \Rightarrow T_b = (200\,\text{K})\left(\frac{7.5\,\text{kPa}}{2.5\,\text{kPa}}\right)\left(\frac{3.0\,\text{m}^3}{1.0\,\text{m}^3}\right)$$

which yields an absolute temperature at b of $T_b = 1.8\times10^3$ K.

(c) As in the previous part:

$$\frac{p_c V_c}{p_a V_a} = \frac{T_c}{T_a} \Rightarrow T_c = (200\,\text{K})\left(\frac{2.5\,\text{kPa}}{2.5\,\text{kPa}}\right)\left(\frac{3.0\,\text{m}^3}{1.0\,\text{m}^3}\right)$$

which yields an absolute temperature at c of $T_c = 6.0\times10^2$ K.

(d) The net energy added to the gas (as heat) is equal to the net work that is done as it progresses through the cycle. The volume change at the largest pressure is an increase. Thus,

$$Q_{\text{net}} = W_{\text{net}} = \frac{1}{2}\left(2.0\,\text{m}^3\right)\left(5.0\times10^3\,\text{Pa}\right) = 5.0\times10^3\ \text{J}.$$

Exercise 7.

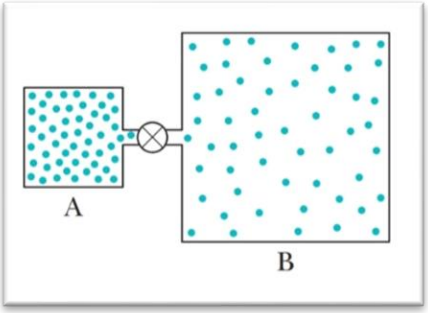
We assume that the pressure of the air in the bubble is essentially the same as the pressure in the surrounding water. If d is the depth of the lake and ρ is the density of water, then the pressure at the bottom of the lake is $p_1 = p_0 + \rho g d$, where p_0 is atmospheric pressure. Since $p_1V_1 = nRT_1$, the number of moles of gas in the bubble is

$$n = p_1V_1/RT_1 = (p_0 + \rho g d)V_1/RT_1.$$

where V_1 is the volume of the bubble at the bottom of the lake and T_1 is the temperature there. At the surface of the lake the pressure is p_0 and the volume of the bubble is $V_2 = nRT_2/p_0$. We substitute for n to obtain

$$\begin{aligned} V_2 &= \frac{T_2}{T_1}\frac{p_0 + \rho g d}{p_0}V_1 \\ &= \left(\frac{293\,\text{K}}{277\,\text{K}}\right)\left(\frac{1.013\times10^5\,\text{Pa} + \left(0.998\times10^3\,\text{kg/m}^3\right)\left(9.8\,\text{m/s}^2\right)(40\,\text{m})}{1.013\times10^5\,\text{Pa}}\right)\left(20\,\text{cm}^3\right) \\ &= 1.0\times10^2\,\text{cm}^3. \end{aligned}$$

Exercise 8.



When the valve is closed the number of moles of the gas in container A is $n_A = p_A V_A/RT_A$ and that in container B is $n_B = 4p_B V_A/RT_B$. The total number of moles in both containers is then

$$n = n_A + n_B = \frac{V_A}{R}\left(\frac{p_A}{T_A} + \frac{4p_B}{T_B}\right) = \text{const.}$$

After the valve is opened, the pressure in container A is $p'_A = Rn'_A T_A/V_A$ and that in container B is $p'_B = Rn'_B T_B/4V_A$. Equating p'_A and p'_B, we obtain $Rn'_A T_A/V_A = Rn'_B T_B/4V_A$, or $n'_B = (4T_A/T_B)n'_A$. Thus,

$$n = n'_A + n'_B = n'_A\left(1 + \frac{4T_A}{T_B}\right) = n_A + n_B = \frac{V_A}{R}\left(\frac{p_A}{T_A} + \frac{4p_B}{T_B}\right).$$

We solve the above equation for n'_A:

$$n'_A = \frac{V}{R}\,\frac{p_A/T_A + 4p_B/T_B}{1 + 4T_A/T_B}.$$

Substituting this expression for n'_A into $p'V_A = n'_A RT_A$, we obtain the final pressure:

$$p' = \frac{n'_A RT_A}{V_A} = \frac{p_A + 4p_B T_A/T_B}{1 + 4T_A/T_B} = 2.0 \times 10^5 \text{ Pa}.$$

Exercise 9.

M = 28.0 g/mol for nitrogen. The rms speed for nitrogen gas at 300 K is 517 m/s.

$$\frac{v_2}{v_1} = \frac{\sqrt{3RT_2/M}}{\sqrt{3RT_1/M}} = \sqrt{\frac{T_2}{T_1}}.$$

(a) With T_2 = 293 K, we obtain v_2 = 511 m/s.

(b) In this case, we set $v_3 = \frac{1}{2}v_2$ and solve $v_3 / v_2 = \sqrt{T3/T2}$ for T_3:

$$T_3 = T_2\left(\frac{v_3}{v_2}\right)^2 = (293\,\text{K})\left(\frac{1}{2}\right)^2 = 73.0\,\text{K}$$

which we write as 73.0 – 273 = – 200 °C.

(c) Now we have $v_4 = 2v_2$ and obtain

$$T_4 = T_2\left(\frac{v_4}{v_2}\right)^2 = (293\,\text{K})(4) = 1.17 \times 10^3\,\text{K}$$

which is equivalent to 899 °C.

Exercise 10.

We can express the ideal gas law in terms of density (ρ =Msam/V) using $n = M_{sam}/M$:

$$pV = \frac{M_{sam}RT}{M} \Rightarrow \rho = \frac{pM}{RT}.$$

We can also use this to write the rms speed formula in terms of density:

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(pM/\rho)}{M}} = \sqrt{\frac{3p}{\rho}}.$$

(a) We convert to SI units: $\rho = 1.24 \times 10^{-2}$ kg/m^3 and $p = 1.01 \times 10^3$ Pa. The rms speed is

$$\sqrt{3(1010)(\frac{kg}{ms2}) / 0.0124\ (\frac{kg}{m})} = 494 \text{ m/s.}$$

(b) We find M from $\rho = pM/RT$ with $T = 273$ K.

$$M = \frac{\rho RT}{p} = \frac{(0.0124\,\text{kg/m}^3)\ \ 8.31\,\text{J/mol}\cdot\text{K}\ \ (273\,\text{K})}{1.01\times10^3\ \text{Pa}} = 0.0279\ \text{kg/mol} = 27.9\ \text{g/mol}.$$

Exercise 11.

(a) Apply the equation for the average translational kinetic energy per molecule of an ideal gas, we yield $K_{avg} = \frac{3}{2}(1.38 \times 10^{-23}\ J/K)(273K) = 5.65 \times 10^{-21}$ J .

(b) For $T = 373$ K, the average translational kinetic energy is $K_{avg} = 7.72 \times 10^{-21}$ J.

(c) The unit mole may be thought of as a (large) collection: 6.02×10^{23} molecules of ideal gas, in this case. Each molecule has energy specified in part (a), so the large collection has a total kinetic energy equal to

K_{mole} (at T= 273K)= $N_A K_{avg} = (6.02 \times 10^{23})(5.65 \times 10^{-21}\ J) = 3.40 \times 10^3$ J.

K_{mole} (at T= 373K)= $N_A K_{avg} = (6.02 \times 10^{23})(7.772 \times 10^{-21}\ J) =$ …..

Exercise 12.

(a) We use $\varepsilon = L_V/n$, where L_V is the heat of vaporization and n is the number of molecules per gram. The molar mass of atomic hydrogen is 1 g/mol and the molar mass of atomic oxygen is 16 g/mol, so the molar mass of H_2O is (1.0 + 1.0 + 16) = 18 g/mol. There are $N_A = 6.02 \times 10^{23}$ molecules in a mole, so the number of molecules in a gram of water is $(6.02 \times 10^{23}\ \text{mol}^{-1})/(18\ \text{g/mol}) = 3.34 \times 10^{22}$ molecules/g. Thus

$$\varepsilon = (539\ \text{cal/g})/(3.34 \times 10^{22}/\text{g}) = 1.61 \times 10^{-20}\ \text{cal} = 6.76 \times 10^{-20}\ \text{J}.$$

(b) The average translational kinetic energy is

$$K_{avg} = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23}\ \text{J/K})[(32.0 + 273.15)\,\text{K}] = 6.32 \times 10^{-21}\ \text{J}.$$

The ratio $\varepsilon/K_{avg} = (6.76 \times 10^{-20}\ J)/(6.32 \times 10^{-21}\ J) = 10.7$.

Exercise 13.

Apply the equation $v = f\lambda$ with v = 331 m/2.

$$f = \frac{v}{\left(\frac{1}{\sqrt{2}\pi d^2 (N/V)}\right)} = (331\,\text{m/s})\,\pi\sqrt{2}\,(3.0\times10^{-10}\,\text{m})^2\left(\frac{nN_A}{V}\right)$$

$$= \left(8.0\times10^7\,\frac{\text{m}^3}{\text{s}\cdot\text{mol}}\right)\left(\frac{n}{V}\right) = \left(8.0\times10^7\,\frac{\text{m}^3}{\text{s}\cdot\text{mol}}\right)\left(\frac{1.01\times10^5\,\text{Pa}}{(8.31\,\text{J/mol}\cdot\text{K})\,(273.15\,\text{K})}\right)$$

$$= 3.5\times10^9\,\text{Hz}$$

where we have used the ideal gas law and substituted $n/V = p/RT$. If we instead use v = 343 m/s (the "default value" for speed of sound in air), then the answer is 3.7 x 10^9 Hz.

Exercise 14.

(a) We use the ideal gas law $pV = nRT = NkT$, where p is the pressure, V is the volume, T is the temperature, n is the number of moles, and N is the number of molecules. The substitutions $N = nN_A$ and $k = R/N_A$ were made. Since 1 cm of mercury = 1333 Pa, the pressure is p = (10–7)(1333 Pa) = 1.333 x 10^{-4} Pa. Thus,

$$\frac{N}{V} = \frac{p}{kT} = \frac{1.333\times10^{-4}\,\text{Pa}}{(1.38\times10^{-23}\,\text{J/K})(295\,\text{K})} = 3.27\times10^{16}\,\text{molecules/m}^3 = 3.27\times10^{10}\,\text{molecules/cm}^3.$$

(b) The molecular diameter is d = 2.00 x 10^{-10} m, the mean free path is

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 N/V} = \frac{1}{\sqrt{2}\pi(2.00\times10^{-10}\,\text{m})^2\,(3.27\times10^{16}\,\text{m}^{-3})} = 172\,\text{m}.$$

Exercise 15.

(a) Utilizing the equation for the mean free path, we set up a ratio:

$$\frac{\lambda_{\text{Ar}}}{\lambda_{\text{N}_2}} = \frac{1/\left(\pi\sqrt{2}d_{\text{Ar}}^2\,(N/V)\right)}{1/\left(\pi\sqrt{2}d_{\text{N}_2}^2\,(N/V)\right)} = \left(\frac{d_{\text{N}_2}}{d_{\text{Ar}}}\right)^2$$

Therefore, we obtain

$$\frac{d_{\text{Ar}}}{d_{\text{N}_2}} = \sqrt{\frac{\lambda_{\text{N}_2}}{\lambda_{\text{Ar}}}} = \sqrt{\frac{27.5\times10^{-6}\ \text{cm}}{9.9\times10^{-6}\ \text{cm}}} = 1.7.$$

(b) Taking into account the equation regarding the Avogadro's number and the ideal gas law, we substitute $N/V = N_An/V = N_Ap/RT$ into the equation for mean free path and find

$$\frac{\lambda_2}{\lambda_1} = \left(\frac{T_2}{T_1}\right)\left(\frac{p_1}{p_2}\right)$$

Comparing (for the same species of molecule) at two different pressures and temperatures,

this leads to

$$\frac{\lambda_2}{\lambda_1} = \left(\frac{T_2}{T_1}\right)\left(\frac{p_1}{p_2}\right)$$

With $\lambda_1 = 9.9 \times 10^{-6}$ cm, $T_1 = 293$ K (the same as T_2 in this part), $p_1 = 750$ torr, and $p_2 =$

150 torr, we find $\lambda_2 = 5.0 \times 10^{-5}$ cm.

(c) The ratio set up in part (b), using the same values for quantities with subscript 1, leads

to $\lambda_2 = 7.9 \times 10^{-6}$ cm for $T_2 = 233$ K and $p_2 = 750$ torr.

Exercise 16.

(a) The average speed is

$$v_{\text{avg}} = \frac{\sum n_i v_i}{\sum n_i} = \frac{[2(1.0)+4(2.0)+6(3.0)+8(4.0)+2(5.0)]\ \text{cm/s}}{2+4+6+8+2} = 3.2\,\text{cm/s}$$

(b) From $v_{\text{rms}} = \sqrt{\sum n_i v_i^2 / \sum n_i}$ we get

$$v_{\text{rms}} = \sqrt{\frac{2(1.0)^2+4(2.0)^2+6(3.0)^2+8(4.0)^2+2(5.0)^2}{2+4+6+8+2}}\ \text{cm/s} = 3.4\,\text{cm/s}.$$

(c) There are eight particles at $v = 4.0$ cm/s, more than the number of particles at any other single speed. So, 4.0 cm/s is the most probable speed.

Exercise 17.

Dividing the Maxwell Speed Distribution (for most probable speed) for the root-mean-square speed of the gas molecules, we find

$$\frac{v_P}{v_{\text{rms}}} = \frac{\sqrt{2RT_2/M}}{\sqrt{3RT_1/M}} = \sqrt{\frac{2T_2}{3T_1}}$$

which, for $v_P = v_{\text{rms}}$, leads to

$$\frac{T_2}{T_1} = \frac{3}{2}\left(\frac{v_P}{v_{\text{rms}}}\right)^2 = \frac{3}{2}$$

MCQs

1. The quantity PV/kT (k = Boltzmann's constant) represents:
a. Number of moles of the gas
b. Total mass of the gas
c. Number of molecules in the gas
d. Density of the gas

Answer: PV/kT = nxN → c)

2. In the process PV = constant, the pressure P versus density graph ρ of an ideal gas is:

a. A parabola
b. A straight line passing through origin
c. A straight line parallel to ρ-axis
d. A straight line parallel to P-axis

Answer: V = nM/ρ, PV = nRT → PM = ρRT → P = ρRT/M → equation of a straight line with slope = RT/M passing through origin

3. The V_{rms} of gas molecules is 300 m/sec. If its absolute temperature is reduced to half and molecular weight is doubled, the V_{rms} will become:

a. 75 m/s
b. 150 m/s
c. 300 m/s
d. 600 m/s

Answer: V_{rms} = rmsq(3RT/M).

V'_{rms} = rmsq(3R1/2T/2M).

→ $V'_{rms} = V_{rms}/2$ → b).

4. Average kinetic energy of molecules is:

a. Directly proportional to square root of temperature

b. Directly proportional to absolute temperature

c. Independent of absolute temperature

d. Inversely proportional to absolute temperature

Answer: b)

5. The specific heat of a gas in an isothermal process is

a. Zero

b. Negative

c. Remains constant

d. Infinite

Answer: d. Q = mcΔT → c = Q/mΔT

6. The specific heat of a gas in an adiabatic process is

a. Zero

b. Negative

c. Remains constant

d. Infinite

Answer: a. $Q = mc\Delta T \rightarrow c = Q/m\Delta T$

7. An ideal gas is taken in a sealed container at 300 K. it is heated at constant volume to a temperature 600 K. the mean kinetic energy of its molecules is

a. Halved

b. Doubled

c. Tripled

d. Quadrupled

Answer: b). KE = 3/2 KT

8. If 70 calories heat is required to raise the temperature of 2 moles of an ideal diatomic gas at constant pressure from 30°C to 35°C. the amount of heat required to raise the temperature of the same gas through the same range at constant volume is

a. 30 cal

b. 50 cal

c. 70 cal

d. 90 cal

Answer: b) $Qp=nCp\Delta T$, $Qv=nCv\Delta T \rightarrow Qv/Qp=Cv/Cp \rightarrow Qv = Cv/Cp \times 70$ cal = 70 cal x 5/7 = 50 cal

9. Which of the following is the unit of specific heat?

a. J kg/°C

b. J/kg°C

c. kg °C/J

d. J kg/$°C^2$

Answer: b)

10. The mean free path of a gas molecule is λ_1 at temperature T. The pressure and temperature of the gas are made two times their original value. The new men free path λ_2 is:

a. Two times of λ_1

b. Equal to λ_1

c. Half of λ_1

d. Four times of λ_1

Answer: $\lambda = \frac{kT}{\sqrt{2}\pi d^2 P}$**,** so **λ** is propotional to T and (1/P). If T and P are both doubled, so **λ**2 = **λ**1**.**